

Full-System Backstepping Landing Control

Mathematical Derivation and Stability Analysis

Contents

1	Plant Description	2
2	Simplifying Assumptions	2
3	Equations of Motion	3
3.1	Full Nonlinear Equations	3
3.2	Small- δ Linearisation (Assumption A4)	3
4	Control Objective and Error Coordinates	4
5	Unified Lyapunov Function — Structure and Motivation	4
6	Step 1 — Virtual Control 1: Desired Pitch ϑ^*	5
7	Step 2 — Virtual Control 2: Desired Angular Rate α_1	6
8	Step 3 — Virtual Control 3: Desired Nozzle Angle α_2	6
9	Step 4 — Real Control: Nozzle Command δ_{cmd}	6
10	Stabilization Proof	7
10.1	Handling the $\alpha_2 \cdot Q$ Residual	7
10.2	Boundedness of All Signals	7
10.3	Convergence of Velocities and Attitude Errors — Barbalat’s Lemma	8
10.4	Convergence of Position Errors — LaSalle’s Invariance Principle	8
10.5	Stability Under Saturation — ISS Analysis	9
10.5.1	Saturation Errors as Bounded Disturbances	9
10.5.2	Perturbed Error Dynamics	9
10.5.3	Lyapunov Derivative Under Disturbances	10
10.5.4	UUB Conclusion	10
10.5.5	Quantitative Bound for the Project Parameters	10
11	Full Algorithm	11
12	Gain Conditions and Tuning Guidelines	12
13	Implementation Notes	13

Abstract

This document derives and formally analyses a **full-system backstepping** landing controller for a planar thrust-vector-controlled (TVC) rocket. A single composite Lyapunov function simultaneously covers position, attitude, and the nozzle actuator. The derivation proceeds in four backstepping steps, producing virtual controls for desired pitch angle, angular rate, and nozzle deflection, before arriving at the real nozzle command. Global practical stability under actuator saturation is proved by establishing that the closed-loop system is **input-to-state stable (ISS)** with respect to actuator clipping errors. ISS is a property we prove for our specific system — not an assumed result — by constructing a Lyapunov inequality of the form $\dot{V} \leq -c\|z\|^2 + D'$, where D' depends on the magnitude of the clipping errors. This guarantees the state remains bounded whenever the disturbances are bounded. The full proof with explicit quantitative bounds is in Section 10.5.

1 Plant Description

The system is a planar TVC rocket in the vertical plane. A single engine deflects by angle δ from the body axis, and δ is governed by a first-order actuator. The rocket operates in the low-speed aerodynamic regime, so aerodynamic drag, normal force, and a pitching moment act on it. All aerodynamic coefficients are assumed known.

State vector and control inputs

$$s = [x, y, \dot{x}, \dot{y}, \vartheta, \dot{\vartheta}, \delta]^T \in \mathbb{R}^7$$

Control inputs: $\sigma \in [\sigma_{\min}, 1]$ (throttle) and $\delta_{\text{cmd}} \in [-\delta_{\max, \text{cmd}}, \delta_{\max, \text{cmd}}]$ (nozzle command).

System parameters

Symbol	Meaning	Units
m	Total mass (constant)	kg
J	Moment of inertia about CoM (constant)	kg m ²
l_{cp}	Distance from CoM to nozzle exit	m
g	Gravitational acceleration, 9.81	m/s ²
$F_{\max}$	Maximum thrust	N
$\delta_{\max}$	Physical nozzle deflection limit	rad
τ_{δ}	Nozzle actuator time constant	s
ρ	Air density, 1.225	kg/m ³
S_m	Reference cross-sectional area	m ²
l	Reference length	m
C_x	Axial drag coefficient, 0.358	—
$C_{y\alpha}$	Normal force coefficient, 0.05403	deg ⁻¹
$C_{m\alpha}$	Pitching moment coefficient, 1.054	rad ⁻¹

2 Simplifying Assumptions

A1. Constant mass. $\dot{m} = 0$, $\dot{J} = 0$, $\dot{l}_{cp} = 0$.

In a real rocket, fuel is consumed during the burn, reducing mass and shifting the centre of mass. Here we assume the landing burn is short enough that mass change is negligible.

A2. Low-speed aerodynamic regime. $V \leq 100$ m/s (Mach < 0.3): $C_x = 0.358$, $C_{m\alpha} = 1.054$ rad $^{-1}$, $C_{y\alpha} = 0.05403$ deg $^{-1}$.

Below Mach 0.3 compressibility effects are small and aerodynamic coefficients are approximately constant.

A3. All parameters known. $C_{m\alpha}$, C_x , $C_{y\alpha}$, J , l_{cp} , τ_δ are all known.

A4. Small nozzle deflection for translational linearisation. $|\delta| \leq \delta_{\max} \leq 0.25$ rad, so:

$$\sin(\vartheta + \delta) \approx \sin \vartheta + \delta \cos \vartheta \quad (1)$$

$$\cos(\vartheta + \delta) \approx \cos \vartheta - \delta \sin \vartheta \quad (2)$$

Error: $|\sin(\vartheta + \delta) - (\sin \vartheta + \delta \cos \vartheta)| \leq \delta^2/2 < 3\%$ for $|\delta| \leq 0.25$ rad.

Because δ is small, the first-order Taylor expansion of $\sin(\vartheta + \delta)$ and $\cos(\vartheta + \delta)$ around $\delta = 0$ is accurate to within 3%. This separates the dominant thrust direction (governed by body pitch ϑ) from the small torque contribution of the nozzle deflection δ , making the equations tractable for backstepping.

A5. Small pitch error for rotational linearisation. $|z_\vartheta| = |\vartheta - \vartheta^*| \ll 1$ rad during the controlled transient, so $\sin(\vartheta^* + z_\vartheta) \approx \sin \vartheta^* + \cos \vartheta^* \cdot z_\vartheta$.

The backstepping design requires writing $\dot{V}_{\text{pos}}$ as an explicit function of the pitch error z_ϑ . The assumption holds whenever the attitude controller is fast enough to keep the tracking error small.

A6. Minimum throttle. $\sigma \geq \sigma_{\min} > 0$, so $g_2 = Fl_{cp}/J > 0$ always.

Enforcing a minimum throttle $\sigma_{\min}$ keeps g_2 bounded away from zero and the controller well-defined at all times.

A7. Full state measurement. $(x, y, \dot{x}, \dot{y}, \vartheta, \dot{\vartheta}, \delta)$ is measurable without noise.

3 Equations of Motion

3.1 Full Nonlinear Equations

The thrust vector makes angle $(\vartheta + \delta)$ from the vertical in the inertial frame. The exact equations of motion are:

$$m\ddot{x} = F \sin(\vartheta + \delta) + X_b \sin \vartheta + Y_b \cos \vartheta \quad (\text{NL1})$$

$$m\ddot{y} = F \cos(\vartheta + \delta) - mg + X_b \cos \vartheta - Y_b \sin \vartheta \quad (\text{NL2})$$

$$J\ddot{\vartheta} = F l_{cp} \sin \delta + C_{m\alpha} \alpha q_\infty S_m l \quad (\text{NL3})$$

$$\tau_\delta \dot{\delta} = -\delta + \delta_{\text{cmd}} \quad (\text{NL4})$$

with $F = \sigma F_{\max}$, $X_b = -C_x q_\infty S_m$, $Y_b = C_{y\alpha} \alpha q_\infty S_m$, $q_\infty = \frac{1}{2} \rho V^2$, $V = \sqrt{\dot{x}^2 + \dot{y}^2}$, and angle of attack:

$$\alpha = \vartheta - \text{atan2}(\dot{x}, \dot{y})$$

3.2 Small- δ Linearisation (Assumption A4)

Applying A4 to (NL1)–(NL2):

$$m\ddot{x} = F \sin \vartheta + F \delta \cos \vartheta + X_b \sin \vartheta + Y_b \cos \vartheta \quad (\text{T1})$$

$$m\ddot{y} = F \cos \vartheta - mg - F \delta \sin \vartheta + X_b \cos \vartheta - Y_b \sin \vartheta \quad (\text{T2})$$

Applying A4 to the rotational equation ($\sin \delta \approx \delta$):

$$J\ddot{\vartheta} = F l_{cp} \delta + C_{m\alpha} \alpha q_\infty S_m l \quad (\text{R1})$$

$$\tau_\delta \dot{\delta} = -\delta + \delta_{\text{cmd}} \quad (\text{A1})$$

Define the shorthand:

$$f_2 = \frac{C_{m\alpha} \alpha q_\infty S_m l}{J} \quad (3)$$

$$g_2 = \frac{F l_{cp}}{J} > 0 \quad (4)$$

so that $\ddot{\vartheta} = g_2 \delta + f_2$. Note that $C_{m\alpha} > 0$ means the rocket body is **aerodynamically unstable** — a positive angle of attack produces a nose-up pitching moment, which amplifies the perturbation. The controller must actively cancel f_2 via the (VC2) term $+\dot{\alpha}_1 - f_2$ in the nozzle reference.

Structural observation. In (T1)–(T2), ϑ drives position through the dominant terms $F \sin \vartheta$, $F \cos \vartheta$, while δ influences position through the smaller coupling terms $F \delta \cos \vartheta$, $-F \delta \sin \vartheta$. Retaining these coupling terms is what makes the full-backstepping approach stronger than a simple outer-inner loop design.

4 Control Objective and Error Coordinates

Landing target:

$$s^* = (x_d, 0, 0, 0, 0, 0, 0)^T$$

Error coordinates:

$$e_x = x - x_d, \quad e_y = y \quad (5)$$

$$\dot{e}_x = \dot{x}, \quad \dot{e}_y = \dot{y} \quad (6)$$

The full objective: $(e_x, e_y, \dot{x}, \dot{y}, \vartheta, \dot{\vartheta}, \delta) \rightarrow 0$ as $t \rightarrow \infty$.

5 Unified Lyapunov Function — Structure and Motivation

The full-backstepping Lyapunov function is built up one term at a time — one $\frac{1}{2}z_i^2$ per step — until the entire state is covered:

$$V = \underbrace{\frac{1}{2}k_{px}e_x^2 + \frac{1}{2}\dot{x}^2 + \frac{1}{2}k_{py}e_y^2 + \frac{1}{2}\dot{y}^2}_{V_{\text{pos}}} + \underbrace{\frac{1}{2}z_\vartheta^2}_{\text{Step 2}} + \underbrace{\frac{1}{2}z_\omega^2}_{\text{Step 3}} + \underbrace{\frac{1}{2}z_\delta^2}_{\text{Step 4}} \quad (\text{V})$$

where the backstepping errors are defined sequentially:

$$z_\vartheta = \vartheta - \vartheta^*, \quad z_\omega = \dot{\vartheta} - \alpha_1, \quad z_\delta = \delta - \alpha_2$$

- z_ϑ : how far the actual pitch ϑ is from the desired pitch ϑ^* computed in Step 1.

- z_ω : how far the actual angular rate $\dot{\vartheta}$ is from the desired rate α_1 computed in Step 2.
- z_δ : how far the actual nozzle angle δ is from the desired nozzle angle α_2 computed in Step 3.

Each virtual control $(\vartheta^*, \alpha_1, \alpha_2)$ is chosen to make the corresponding new term in $\dot{V}$ negative-definite, while passing a residual cross-coupling to the next step.

6 Step 1 — Virtual Control 1: Desired Pitch ϑ^*

Compute $\dot{V}_{\text{pos}}$:

$$\dot{V}_{\text{pos}} = k_{px}e_x\dot{x} + \dot{x}\ddot{x} + k_{py}e_y\dot{y} + \dot{y}\ddot{y}$$

Choose the desired accelerations:

$$\begin{aligned} A_x &= -k_{px}e_x - k_{dx}\dot{x} \\ A_y &= -k_{py}e_y - k_{dy}\dot{y} \end{aligned} \tag{O1}$$

and require $(F/m)\sin\vartheta^* = A_x$, $(F/m)\cos\vartheta^* = A_y + g$. This gives:

Remark 6.1 (Aerodynamics omitted from feedforward). The desired accelerations (O1) account only for thrust, not for aerodynamic forces X_b, Y_b . Including them would make the computation circular: computing ϑ^* requires knowing X_b, Y_b , which depend on the angle of attack $\alpha = \vartheta - \text{atan2}(\dot{x}, \dot{y})$, which in turn depends on ϑ^* . Instead, aerodynamic forces are treated as bounded disturbances absorbed into the ISS bound D of Section 10.5. This is justified because during the landing burn thrust dominates aerodynamics by roughly an order of magnitude.

$$\sigma = \text{clip}\left(\frac{m\sqrt{A_x^2 + (A_y + g)^2}}{F_{\text{max}}}, \sigma_{\text{min}}, 1\right) \tag{O2}$$

$$\vartheta^* = \text{atan2}(A_x, A_y + g) \tag{O3}$$

Hover check: $e_x = e_y = \dot{x} = \dot{y} = 0 \Rightarrow A_x = 0, A_y = 0, \sigma = mg/F_{\text{max}}, \vartheta^* = 0$. ✓

Now substitute (T1)–(T2) and (O1) into $\dot{V}_{\text{pos}}$, applying A5: $\sin\vartheta \approx \sin\vartheta^* + \cos\vartheta^* \cdot z_\vartheta$. Using $(F/m)\sin\vartheta^* = A_x$ and $(F/m)\cos\vartheta^* = A_y + g$.

Derivation of the x -channel term. Substitute $\ddot{x}$ from (T1):

$$\dot{x}\ddot{x} + k_{px}e_x\dot{x} = \dot{x}\left(\frac{F\sin\vartheta + F\delta\cos\vartheta + X_b\sin\vartheta + Y_b\cos\vartheta}{m} + k_{px}e_x\right)$$

Apply A5 to expand $F\sin\vartheta/m$ around ϑ^* :

$$\frac{F\sin\vartheta}{m} \approx A_x + \frac{F\cos\vartheta^*}{m}z_\vartheta$$

With $A_x = -k_{px}e_x - k_{dx}\dot{x}$ from (O1), the first group gives $-k_{dx}\dot{x}^2$. The aerodynamic normal-force contribution Y_bV_n/m is not sign-definite in general; it is absorbed into the ISS bound D as an additional bounded term $\bar{d}_Y$. Collecting the sign-definite terms:

$$\dot{V}_{\text{pos}} \leq -k_{dx}\dot{x}^2 - k_{dy}\dot{y}^2 + z_\vartheta \underbrace{\frac{F}{m}(\dot{x}\cos\vartheta^* - \dot{y}\sin\vartheta^*)}_{\triangleq P} + \delta \underbrace{\frac{F}{m}(\dot{x}\cos\vartheta - \dot{y}\sin\vartheta)}_{\triangleq Q} + \bar{d}_Y \tag{DV1}$$

where P is the **position-attitude coupling** and Q is the **nozzle-position coupling**.

7 Step 2 — Virtual Control 2: Desired Angular Rate α_1

Augment the Lyapunov function: $V_2 = V_{\text{pos}} + \frac{1}{2}z_{\vartheta}^2$.

Compute $\dot{z}_{\vartheta}$: $\dot{z}_{\vartheta} = \dot{\vartheta} - \dot{\vartheta}^*$.

Using $\dot{\vartheta} = \alpha_1 + z_{\omega}$ and grouping the z_{ϑ} contributions:

$$\dot{V}_2 = -k_{dx}\dot{x}^2 - k_{dy}\dot{y}^2 + z_{\vartheta}(P + \alpha_1 + z_{\omega} - \dot{\vartheta}^*) + \delta \cdot Q + \bar{d}_Y$$

Design step. Require $P + \alpha_1 - \dot{\vartheta}^* = -k_{\vartheta}z_{\vartheta}$:

$$\boxed{\alpha_1 = \dot{\vartheta}^* - k_{\vartheta}z_{\vartheta} - \frac{F}{m}(\dot{x} \cos \vartheta^* - \dot{y} \sin \vartheta^*)} \quad (\text{VC1})$$

Then $z_{\vartheta}(P + \alpha_1 + z_{\omega} - \dot{\vartheta}^*) = -k_{\vartheta}z_{\vartheta}^2 + z_{\vartheta}z_{\omega}$.

Result after Step 2:

$$\dot{V}_2 = -k_{dx}\dot{x}^2 - k_{dy}\dot{y}^2 - k_{\vartheta}z_{\vartheta}^2 + z_{\vartheta}z_{\omega} + \delta \cdot Q + \bar{d}_Y \quad (\text{DV2})$$

8 Step 3 — Virtual Control 3: Desired Nozzle Angle α_2

Augment: $V_3 = V_2 + \frac{1}{2}z_{\omega}^2$.

Compute $\dot{z}_{\omega}$: $\dot{z}_{\omega} = \ddot{\vartheta} - \dot{\alpha}_1 = g_2\delta + f_2 - \dot{\alpha}_1$.

Grouping all δ -dependent terms, with $\Gamma \triangleq g_2z_{\omega} + Q$:

$$\dot{V}_3 = -k_{dx}\dot{x}^2 - k_{dy}\dot{y}^2 - k_{\vartheta}z_{\vartheta}^2 + z_{\omega}(z_{\vartheta} + f_2 - \dot{\alpha}_1) + \delta \cdot \Gamma + \bar{d}_Y \quad (\text{DV3})$$

Design step. Write $\delta = \alpha_2 + z_{\delta}$ and choose α_2 so that the z_{ω} bracket gives $-k_{\omega}z_{\omega}^2$:

$$z_{\omega}(z_{\vartheta} + f_2 - \dot{\alpha}_1 + g_2\alpha_2) = -k_{\omega}z_{\omega}^2$$

$$\boxed{\alpha_2 = \frac{1}{g_2}(-k_{\omega}z_{\omega} - z_{\vartheta} - f_2 + \dot{\alpha}_1)} \quad (\text{VC2})$$

This requires $g_2 \neq 0$ — guaranteed by Assumption A6.

Result after Step 3:

$$\dot{V}_3 = -k_{dx}\dot{x}^2 - k_{dy}\dot{y}^2 - k_{\vartheta}z_{\vartheta}^2 - k_{\omega}z_{\omega}^2 + z_{\delta} \cdot \Gamma + \alpha_2 \cdot Q + \bar{d}_Y \quad (\text{DV3}')$$

9 Step 4 — Real Control: Nozzle Command δ_{cmd}

Augment:

$$V = V_3 + \frac{1}{2}z_{\delta}^2 = \frac{1}{2}k_{px}e_x^2 + \frac{1}{2}\dot{x}^2 + \frac{1}{2}k_{py}e_y^2 + \frac{1}{2}\dot{y}^2 + \frac{1}{2}z_{\vartheta}^2 + \frac{1}{2}z_{\omega}^2 + \frac{1}{2}z_{\delta}^2 \quad (\text{V4})$$

Compute $\dot{z}_{\delta}$:

$$\dot{z}_{\delta} = \dot{\delta} - \dot{\alpha}_2 = \frac{-\delta + \delta_{\text{cmd}}}{\tau_{\delta}} - \dot{\alpha}_2 \quad (\text{D4})$$

Design step. Choose δ_{cmd} to make the bracket multiplying z_δ equal to $-k_\delta z_\delta$:

$$\boxed{\delta_{\text{cmd}} = \delta + \tau_\delta(\dot{\alpha}_2 - g_2 z_\omega - Q - k_\delta z_\delta)} \quad (\text{CL})$$

$\dot{V}$ after substituting (CL):

$$\dot{V} = -k_{dx}\dot{x}^2 - k_{dy}\dot{y}^2 - k_\vartheta z_\vartheta^2 - k_\omega z_\omega^2 - k_\delta z_\delta^2 + \alpha_2 \cdot Q + \bar{d}_Y \quad (\text{ISS1})$$

10 Stabilization Proof

10.1 Handling the $\alpha_2 \cdot Q$ Residual

The coupling $Q = (F/m)(\dot{x} \cos \vartheta - \dot{y} \sin \vartheta)$ satisfies $|Q| \leq (F/m)\sqrt{\dot{x}^2 + \dot{y}^2}$. Applying Young's inequality with parameter $\epsilon > 0$:

$$|\alpha_2 \cdot Q| \leq \frac{\epsilon}{2} \alpha_2^2 + \frac{(F/m)^2}{2\epsilon} (\dot{x}^2 + \dot{y}^2)$$

Choosing ϵ such that:

$$\epsilon \geq \frac{(F/m)^2}{2 \min(k_{dx}, k_{dy})} \quad (\text{C-eps})$$

the Q^2 residual is absorbed by the velocity damping terms, and (ISS1) becomes:

$$\dot{V} \leq -k'_{dx} \dot{x}^2 - k'_{dy} \dot{y}^2 - k_\vartheta z_\vartheta^2 - k_\omega z_\omega^2 - k_\delta z_\delta^2 + \frac{\epsilon}{2} \alpha_2^2 \quad (\text{ISS2})$$

where $k'_{dx} = k_{dx} - (F/m)^2/(2\epsilon) > 0$ and $k'_{dy} = k_{dy} - (F/m)^2/(2\epsilon) > 0$.

10.2 Boundedness of All Signals

Note on the $\frac{\epsilon}{2} \alpha_2^2$ term. From (VC2), $\alpha_2 = \frac{1}{g_2}(-k_\omega z_\omega - z_\vartheta - f_2 + \dot{\alpha}_1)$, so α_2 is a function of z_ω , z_ϑ , f_2 , and $\dot{\alpha}_1$, all covered by V . Since $V \geq \frac{1}{2} z_\vartheta^2$ and $V \geq \frac{1}{2} z_\omega^2$:

$$|z_\vartheta| \leq \sqrt{2V} \quad (7)$$

$$|z_\omega| \leq \sqrt{2V} \quad (8)$$

Similarly $|f_2|$ is bounded (depends on bounded airspeed and angle of attack), and $|\dot{\alpha}_1|$ is bounded as a continuous function of bounded signals. Combining, there exists a constant C_{α_2} depending only on $V(0)$ and the gains such that:

$$\frac{\epsilon}{2} \alpha_2^2 \leq \frac{\epsilon}{2} C_{\alpha_2}^2 \triangleq \mu$$

Substituting into (ISS2):

$$\dot{V} \leq -k'_{dx} \dot{x}^2 - k'_{dy} \dot{y}^2 - k_\vartheta z_\vartheta^2 - k_\omega z_\omega^2 - k_\delta z_\delta^2 + \mu \quad (\text{ISS3})$$

Theorem 10.1 (UUB). *Under the control law (CL), virtual controls (VC1)–(VC2), and Assumptions A1–A7, if gains satisfy (C-eps) strictly, then all signals $(e_x, \dot{x}, e_y, \dot{y}, z_\vartheta, z_\omega, z_\delta)$ are **uniformly ultimately bounded**: they enter and remain in the compact set $\Omega = \{V \leq \mu/c\}$ in finite time, where:*

$$c = \min(k'_{dx}, k'_{dy}, k_\vartheta, k_\omega, k_\delta)$$

The transient bound is $V(t) \leq \max(V(0), \mu/c) \triangleq V_{\max}$. Individual bounds:

$$|e_x(t)| \leq \sqrt{\frac{2V_{\max}}{k_{px}}} \triangleq B_{e_x} \quad (9)$$

$$|\dot{x}(t)| \leq \sqrt{2V_{\max}} \triangleq B_{\dot{x}} \quad (10)$$

and analogously for $e_y, \dot{y}, z_\theta, z_\omega, z_\delta$. All bounds depend only on $V(0)$, μ , and the gains.

Remark 10.1 (Two-phase stability picture). The overall stability result has two phases:

- **Phase 1 — transient** ($\alpha_2 \neq 0, \mu > 0$): While the system is away from equilibrium, $\alpha_2 \neq 0$ and the residual $\mu = \frac{\epsilon}{2}C_{\alpha_2}^2 > 0$ from the $\alpha_2 \cdot Q$ cross-term is nonzero. Additionally, clipping of $\alpha_1, \alpha_2, \delta_{\text{cmd}}$ may be active, contributing the extra disturbance D' of Section 10.5. In this phase only UUB holds.
- **Phase 2 — steady descent** ($\alpha_2 \rightarrow 0, \mu \rightarrow 0$): As the system approaches equilibrium, $\alpha_2 \rightarrow 0$ so $\mu \rightarrow 0$, and clipping becomes inactive. The stronger results of Sections 10.3 and 10.4 apply.

10.3 Convergence of Velocities and Attitude Errors — Barbalat's Lemma

Remark 10.2 (Scope). The argument below assumes $\mu = 0$ — i.e., the $\alpha_2 \cdot Q$ cross-term is neglected. This is a standard idealised analysis. The true result is UUB (Section 10.2). The argument shows what the system would do in the absence of the residual, and μ is small when α_2 is small.

Theorem 10.2 (Idealised, $\mu = 0$). *If the $\alpha_2 \cdot Q$ residual is neglected, then $\dot{x}(t) \rightarrow 0, \dot{y}(t) \rightarrow 0, z_\theta(t) \rightarrow 0, z_\omega(t) \rightarrow 0, z_\delta(t) \rightarrow 0$ as $t \rightarrow \infty$.*

Proof. We show the argument for $\dot{x}$; the others follow identically.

Step 1 — Integrability. With $\mu = 0$, (ISS3) gives $\dot{V} \leq -k'_{dx} \dot{x}^2$. Integrating from 0 to T :

$$\int_0^T k'_{dx} \dot{x}^2(\tau) d\tau \leq V(0) - V(T) \leq V(0) < \infty$$

Taking $T \rightarrow \infty$: $\dot{x}^2 \in L^2([0, \infty))$.

Step 2 — Uniform continuity. From Section 10.2, $|\dot{x}| \leq \sqrt{2V_{\max}}$. From (T1), $\ddot{x}$ is a continuous function of bounded signals, so $\frac{d}{dt}(\dot{x}^2) = 2\dot{x}\ddot{x}$ is bounded, meaning $\dot{x}^2$ is uniformly continuous.

Step 3 — Barbalat's Lemma. A non-negative uniformly continuous function with finite L^2 integral converges to zero:

$$\dot{x}^2(t) \rightarrow 0 \implies \dot{x}(t) \rightarrow 0 \quad \text{as } t \rightarrow \infty. \quad \square$$

□

10.4 Convergence of Position Errors — LaSalle's Invariance Principle

Theorem 10.3 (Idealised, same scope as Section 10.3). *If $\mu = 0$, then $e_x(t) \rightarrow 0$ and $e_y(t) \rightarrow 0$ as $t \rightarrow \infty$.*

Proof. The closed-loop system is **autonomous** — the right-hand side depends only on the state, not explicitly on t . LaSalle's invariance principle (Khalil, Theorem 4.4) therefore applies.

Define the limiting set:

$$\mathcal{S} = \{(e_x, \dot{x}, e_y, \dot{y}, z_\theta, z_\omega, z_\delta) : \dot{x} = \dot{y} = z_\theta = z_\omega = z_\delta = 0\}$$

In $\mathcal{S}$: $\dot{x} = \dot{y} = 0 \Rightarrow \ddot{x} = \ddot{y} = 0$; $Q = 0$; $z_\vartheta = 0 \Rightarrow \vartheta = \vartheta^*$; $z_\delta = 0 \Rightarrow \delta = \alpha_2$. Substituting $\dot{x} = \dot{y} = 0$ into (T1) (aerodynamic terms vanish since $V = 0$):

$$m\ddot{x} = F \sin \vartheta^*$$

By construction (O3), $F \sin \vartheta^* = mA_x$, and with $\ddot{x} = 0$:

$$0 = A_x = -k_{px}e_x - k_{dx}\underbrace{\dot{x}}_{=0} = -k_{px}e_x \implies e_x = 0$$

The same argument applied to (T2) gives $e_y = 0$.

Therefore the largest invariant subset of $\mathcal{S}$ is $\{e_x = e_y = 0\}$, and $(e_x(t), e_y(t)) \rightarrow 0$. $\square$ $\square$

10.5 Stability Under Saturation — ISS Analysis

Section 10.2 (UUB) holds regardless of clipping. The idealised convergence proofs in Sections 10.3–10.4 additionally assume that α_1 , α_2 , and δ_{cmd} are delivered exactly. In the real implementation all three are clipped:

$$\alpha_{1,\text{sat}} = \text{clip}(\alpha_{1,\text{raw}}, -\alpha_{1,\text{max}}, \alpha_{1,\text{max}}) \quad (11)$$

$$\alpha_{2,\text{sat}} = \text{clip}(\alpha_{2,\text{raw}}, -\delta_{\text{max}}, \delta_{\text{max}}) \quad (12)$$

$$\delta_{\text{cmd,sat}} = \text{clip}(\delta_{\text{cmd}}, -\delta_{\text{max,cmd}}, \delta_{\text{max,cmd}}) \quad (13)$$

This section proves that the closed-loop system remains **input-to-state stable (ISS)** with respect to the saturation errors.

10.5.1 Saturation Errors as Bounded Disturbances

Each saturation is modelled as an additive bounded disturbance.

α_1 **saturation:**

$$d_\vartheta = \alpha_{1,\text{sat}} - \alpha_{1,\text{raw}}, \quad |d_\vartheta| \leq 2\alpha_{1,\text{max}} \triangleq \bar{d}_\vartheta \quad (14)$$

α_2 **saturation:**

$$d_\omega = \alpha_{2,\text{sat}} - \alpha_{2,\text{raw}}, \quad |d_\omega| \leq 2\delta_{\text{max}}, \quad |g_2 d_\omega| \leq g_{2,\text{max}} \cdot 2\delta_{\text{max}} \triangleq \bar{d}_\omega \quad (15)$$

δ_{cmd} **saturation:**

$$d_\delta = \delta_{\text{cmd,sat}} - \delta_{\text{cmd}}, \quad |d_\delta| \leq 2\delta_{\text{max,cmd}}, \quad |d_\delta/\tau_\delta| \leq 2\delta_{\text{max,cmd}}/\tau_\delta \triangleq \bar{d}_\delta \quad (16)$$

10.5.2 Perturbed Error Dynamics

Substituting $\alpha_{2,\text{sat}} = \alpha_{2,\text{raw}} + d_\omega$ and using the design choice (VC2):

$$\dot{z}_\omega = -k_\omega z_\omega - z_\vartheta + g_2 z_\delta + g_2 d_\omega - d_\vartheta$$

Substituting $\delta_{\text{cmd,sat}} = \delta_{\text{cmd}} + d_\delta$ and using (CL):

$$\dot{z}_\delta = -k_\delta z_\delta - g_2 z_\omega - Q + \frac{d_\delta}{\tau_\delta}$$

10.5.3 Lyapunov Derivative Under Disturbances

The $g_2 z_\omega z_\delta$ terms cancel exactly. Bounding the remaining cross-terms via Young's inequality with $\lambda_3 = k_\delta/4$, $\lambda_4 = k_\vartheta/2$, $\lambda_1 = \lambda_5 = k_\omega/4$, $\lambda_2 = k_\delta/8$:

$$\dot{V} \leq -k''_{dx} \dot{x}^2 - k''_{dy} \dot{y}^2 - \frac{k_\vartheta}{2} z_\vartheta^2 - \frac{k_\omega}{4} z_\omega^2 - \frac{k_\delta}{8} z_\delta^2 + D' \quad (\text{ISS4})$$

where the total disturbance bound is:

$$D' = \mu + \frac{2g_2^2 \bar{d}_\omega^2}{k_\omega} + \frac{2\bar{d}_\vartheta^2}{k_\omega} + \frac{4\bar{d}_\delta^2}{k_\delta \tau_\delta^2} \quad (17)$$

10.5.4 UUB Conclusion

Theorem 10.4 (UUB under saturation). *Under Assumptions A1–A7, gain condition (C-eps), and $k_\omega > 1/k_\vartheta + 1$, the closed-loop system with saturated controls is **uniformly ultimately bounded**: all states enter and remain in $\mathcal{B} = \{V \leq D/c\}$, where:*

$$c = \min\left(k''_{dx}, k''_{dy}, \frac{k_\vartheta}{2}, \frac{k_\omega}{4}, \frac{k_\delta}{8}\right) \quad (18)$$

$$D = D' \quad (19)$$

The residual bounds on attitude errors are:

$$|z_\omega|_\infty \leq \sqrt{\frac{2D}{c k_\omega}} \quad (20)$$

$$|z_\delta|_\infty \leq \sqrt{\frac{2D}{c k_\delta}} \quad (21)$$

10.5.5 Quantitative Bound for the Project Parameters

With parameters from `configs/backstepping.yaml`:

- $g_{2,\max} = F_{\max} l_{cp}/J = 191763 \times 10.5/318196.65 \approx 6.33 \text{ rad/s}^2/\text{rad}$
- $\alpha_{1,\max} = 1.5 \text{ rad/s}$, so $\bar{d}_\vartheta = 3.0 \text{ rad/s}$
- $\delta_{\max} = 15 = 0.262 \text{ rad}$, so $\bar{d}_\omega = 2 \times 6.33 \times 0.262 \approx 3.31 \text{ rad/s}^2$
- $\delta_{\max,\text{cmd}} = 0.262 \text{ rad}$, $\tau_\delta = 0.05 \text{ s}$, so $\bar{d}_\delta \approx 10.47 \text{ rad/s}$

With $k_\omega = 5$, $k_\delta = 15$, $\tau_\delta = 0.05 \text{ s}$, the three contributions to D' are:

$$\frac{2g_{2,\max}^2 \bar{d}_\omega^2}{k_\omega} \approx 175.8 \quad (22)$$

$$\frac{2\bar{d}_\vartheta^2}{k_\omega} \approx 3.6 \quad (23)$$

$$\frac{4\bar{d}_\delta^2}{k_\delta \tau_\delta^2} \approx 11697 \quad (24)$$

The δ_{cmd} clipping term dominates — it is amplified by $1/\tau_\delta^2 = 400$. Total $D' \approx \mu + 11876$.

With $c = \min(k_\vartheta/2, k_\omega/4, k_\delta/8) = \min(3.0, 1.25, 1.875) = 1.25$:

$$|z_\omega|_\infty \leq \sqrt{\frac{2 \times 11876}{1.25 \times 5}} \approx 61.6 \text{ rad/s} \approx 3532/\text{s}$$

This large bound assumes all three clipping saturations are **permanently active simultaneously**. In the simulation, α_1 clipping clears within 0.1 s and α_2 clipping within 1–2 s; after that $d_\vartheta = d_\omega = d_\delta = 0$ exactly, $\mu \rightarrow 0$, and the results of Sections 10.3–10.4 apply.

11 Full Algorithm

State: $s = [x, y, \dot{x}, \dot{y}, \vartheta, \dot{\vartheta}, \delta]^T$. Filter state: α_2^f . Derivative filter states: $\dot{\vartheta}_{\text{filt}}^*$, $\dot{\alpha}_{1,\text{filt}}$.

Block A — Step 1: Virtual control 1 (ϑ^* , σ)

```

1.  e_x = x - x_d,    e_y = y
2.  A_x = -k_px * e_x - k_dx * xdot
3.  A_y = -k_py * e_y - k_dy * ydot
4.  a_vert = max(A_y + g, |A_x| / tan(phi_lim), a_min)    [safety clamp]
5.  A_y = a_vert - g
6.  A_x = clip(A_x, -a_vert*tan(phi_lim), a_vert*tan(phi_lim))
7.  T_des = sqrt(A_x^2 + a_vert^2)
8.  sigma = clip(m * T_des / F_max, sigma_min, 1)
9.  F = sigma * F_max
10. theta* = atan2(A_x, a_vert)
11. theta*_dot ~= filtered analytically (see Section 13)

```

Remark 11.1 (Safety clamping, steps 4–6). The theoretical design requires only $a_{\min} = 0$ for the Lyapunov proof; the implementation enforces $a_{\min} = 0.5 \text{ m/s}^2$ and the tilt limit $\phi_{\text{lim}} = 30$ to keep the rocket from commanding full free-fall or excessive tilt. The clamp on A_x ensures $|\vartheta^*| \leq \phi_{\text{lim}}$ exactly. These modifications introduce a bounded error in (O1) handled by Section 10.5.

Block B — Aerodynamics

```

12. V_spd = sqrt(xdot^2 + ydot^2)
13. alpha_aoa = (V_spd < V_min) ? theta : theta - atan2(xdot, ydot)
14. q_inf = 0.5 * rho * V_spd^2
15. f2 = C_ma * alpha_aoa * q_inf * S_m * l / J
16. g2 = F * l_cp / J

```

Block C — Step 2: Virtual control 2 (α_1)

```

17. z_theta = theta - theta*
18. alpha1 = theta*_dot - k_phi * z_theta
19. z_omega = theta_dot - alpha1
20. alpha1_dot ~= (alpha1(t) - alpha1(t-dt)) / dt

```

Remark 11.2 (P-coupling omission). The theoretical formula (VC1) includes $-(F/m)(\dot{x} \cos \vartheta^* - \dot{y} \sin \vartheta^*)$. In practice this reaches 60+ rad/s at high entry velocities and permanently saturates α_1 . The simplified form $\alpha_1 = \dot{\vartheta}^* - k_\vartheta z_\vartheta$ is used instead. The omitted P term is a bounded disturbance covered by Section 10.5.

Remark 11.3 (Filtered derivatives). The implementation propagates $\dot{\vartheta}^*$ and $\dot{\alpha}_1$ as ODE states driven by first-order filters toward their instantaneous analytical values (bandwidth $N = 50$ rad/s). The lag error is bounded by $O(1/N)$ and constitutes a bounded disturbance within the ISS framework.

Remark 11.4 (α_1 saturation). The implementation clips α_1 to $\pm\alpha_{1,\text{max}}$ (default 1.5 rad/s). This introduces a bounded disturbance in the z_ω dynamics analogous to the α_2 saturation analysed in Section 10.5, with $\bar{d}_{\alpha_1} = 2\alpha_{1,\text{max}}$.

Block D — Step 3: Virtual control 3 (α_2)

```

21. alpha2_raw = (1/g2) * (-k_omega * z_omega - z_theta - f2 +
    alpha1_dot)

```

```
22. alpha2_sat = clip(alpha2_raw, -delta_max, delta_max)
```

Block E — Command filter

```
23. alpha2f_dot = (alpha2_sat - alpha2f) / tau_f
24. alpha2f     += alpha2f_dot * dt      [or integrate inside ODE]
25. z_delta     = delta - alpha2f
```

Remark 11.5 (Command filtering). The Lyapunov derivation defines $z_\delta = \delta - \alpha_2$ and uses $\dot{\alpha}_2$ in (CL). In the implementation, α_2 is replaced by its filtered version α_2^f , so $z_\delta = \delta - \alpha_2^f$. Command filtering (Farrell et al., 2005) avoids differentiating α_2 analytically; the filter error introduces an additional bounded disturbance of order $O(\tau_f)$ absorbed by Section 10.5.

Block F — Step 4: Real control (δ_{cmd})

```
26. delta_cmd = delta + tau_delta * (alpha2f_dot - g2*z_omega - k_delta*
    z_delta)
27. delta_cmd = clip(delta_cmd, -delta_max_cmd, delta_max_cmd)
```

Remark 11.6 (Q-coupling omission). The theoretical formula (CL) includes $-Q = -(F/m)(\dot{x} \cos \vartheta - \dot{y} \sin \vartheta)$. This term reaches 50+ rad/s at high entry velocities and, multiplied by τ_δ , produces nozzle commands far exceeding $\delta_{\text{max,cmd}}$. The term is omitted in the implementation; the bounded error is covered by Section 10.5.

Output: $[\sigma, \delta_{\text{cmd}}]$

12 Gain Conditions and Tuning Guidelines

Necessary conditions

$$k_{px}, k_{dx}, k_{py}, k_{dy}, k_\vartheta, k_\omega, k_\delta > 0, \quad \sigma_{\min} > 0, \quad \tau_f > 0$$

Critical damping of position loop

$$k_{dx} \geq 2\sqrt{k_{px}} \tag{25}$$

$$k_{dy} \geq 2\sqrt{k_{py}} \tag{26}$$

Attitude loop bandwidth

A practical guideline (no formal time-scale separation required):

$$\sqrt{k_\vartheta g_{2,\text{hov}}} \geq 3\sqrt{k_{px}} \tag{27}$$

$$g_{2,\text{hov}} = \frac{\sigma_{\text{hov}} F_{\text{max}} l_{cp}}{J} \tag{28}$$

Cross-term damping condition

From the $z_\vartheta z_\omega$ bounding step in Section 10.5:

$$k_\omega > \frac{1}{k_\vartheta} + 1$$

For the project gains $k_\vartheta = 6$: $k_\omega > 1/6 + 1 \approx 1.17$. Current value $k_\omega = 5$ satisfies this with large margin.

Residual $\alpha_2 \cdot Q$ condition (C-eps)

$$\frac{(F/m)^2}{2\epsilon} \leq \min(k_{dx}, k_{dy}) \iff \epsilon \geq \frac{(F/m)^2}{2\min(k_{dx}, k_{dy})}$$

Since ϵ is a free parameter in Young's inequality (not a gain to tune), this is always satisfiable.

Command filter time constant

$$\frac{\tau_\delta}{10} \lesssim \tau_f \lesssim \frac{\tau_\delta}{5}$$

13 Implementation Notes

Ten-state ODE

```
state = [x, y, vx, vy, phi, omega, delta, alpha2f, ts_dot, a1_dot]
```

The last two states are first-order filtered derivatives: `ts_dot` tracks ϑ^* and `a1_dot` tracks $\dot{\alpha}_1$, driven analytically inside the ODE.

Filtered derivatives

```
ts_dot_instant = ((A_y+g)*dAx_dt - A_x*dAy_dt) / max(A_x**2+(A_y+g)**2,
    a_vert**2)
ts_dot_instant = clip(ts_dot_instant, -alpha1_max, alpha1_max)
a1_dot_instant = ts_dot_instant - k_phi * (omega - ts_dot)

d/dt(ts_dot) = N_deriv * (ts_dot_instant - ts_dot)    # N_deriv = 50 rad/s
d/dt(a1_dot) = N_deriv * (a1_dot_instant - a1_dot)
```

Python variable mapping

Symbol	Python variable
ϑ	<code>phi</code>
$\dot{\vartheta}$	<code>omega</code>
ϑ^*	<code>theta_star</code>
δ	<code>delta</code>
δ_{cmd}	<code>delta_cmd</code>
α_2^f	<code>alpha2f</code>
$z_\vartheta, z_\omega, z_\delta$	<code>z_phi, z_omega, z_delta</code>
α_1, α_2	<code>alpha1, alpha2</code>
P, Q	<code>P_coupling, Q_coupling</code>
f_2, g_2	<code>f2, g2</code>
σ	<code>sigma</code>
$k_\vartheta, k_\omega, k_\delta$	<code>k_phi, k_omega, k_delta</code>
$k_{px}, k_{dx}, k_{py}, k_{dy}$	<code>k_px, k_dx, k_py, k_dy</code>
τ_δ, τ_f	<code>tau_delta, tau_filter</code>

References

1. Khalil, H. K. (2002). *Nonlinear Systems* (3rd ed.). Prentice Hall.

2. Krstić, M., Kanellakopoulos, I., & Kokotović, P. (1995). *Nonlinear and Adaptive Control Design*. Wiley.
3. Farrell, J. A., Sharma, M., & Polycarpou, M. (2005). Backstepping-based flight control with adaptive function approximation. *AIAA Journal of Guidance, Control, and Dynamics*, 28(6), 1089–1102.
4. Slotine, J.-J.E., & Li, W. (1991). *Applied Nonlinear Control*. Prentice Hall.
5. Wie, B. (1998). *Space Vehicle Dynamics and Control*. AIAA.
6. Sontag, E. D. (1989). Smooth stabilization implies coprime factorization. *IEEE Trans. Autom. Control*, 34(4), 435–443.
7. Sontag, E. D., & Wang, Y. (1995). On characterizations of the input-to-state stability property. *Systems & Control Letters*, 24(5), 351–359.